HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LA

Life Insurance Mathematics A

Semester 1 2017/18

Duration: Two Hours

Please answer all 7 questions.

Total marks: 60

Electronic calculators may be used, provided that they conform to University regulations.

You should have a copy of the yellow Formulae and Tables for Actuarial Examinations on your desk.

Question 1

Suppose that, for integer values of x between 50 and 60 inclusive,

$p_x = (12{,}400 - x^2) / 10{,}000$

Suppose also that the effective rate of interest is 9% per year.

Calculate the following:

(a) q_{53} [1 mark]

(b) ${}_3p_{56}$ [2 marks]

(c) The expected present value of a term assurance for a life aged 55 with a term of 2 years and a sum assured of $1 million, payable at the end of the year of death. [3 marks]

(d) The standard deviation of the present value in (c). [3 marks]

[Total 9 marks]

Question 2

When a woman was aged 30 exact, a life office sold her a term assurance with a fixed sum assured of £100,000 and a term of 20 years. The benefit is payable at the end of the year of death. The premium was fixed at £300 per year, payable annually in advance. She is now aged 48 exact.

Calculate the gross premium reserve, just before the premium payable at age 48, on the following basis.

Mortality: $q_{48} = 0.012$, and $q_{49} = 0.013$

Rate of interest: 2% per year

Premium expenses: 2% of each premium

Claim expenses: 0.2% of sum assured [Total 7 marks]

[**PLEASE TURN OVER]**

Question 3

(a) Write the formula for the force of mortality, μ_x . [2 marks]

(b) Calculate the probability that a life aged 0 will die between ages 15 and 39, given the following survival function:

$S_0(x) = 1/8\ (64 - x)^{0.5}$. [3 marks]

[Total 5 marks]

Question 4

On 1 January 2005 a life insurance company issued a number of pure endowment policies with a term of 20 years to a group of lives aged 45 exact. In each case, the sum assured was £200,000. Premiums were payable annually in advance throughout the term, or until earlier death. On 1 January 2016, 17,000 policies were still in force. During 2016, 103 policyholders died, and no policy lapsed for any other reason.

(a) Calculate the annual premium. [3 marks]

(b) Calculate the mortality profit or loss for this group for the year ending 31 December 2016. [7 marks]

The same basis was used for the premium and reserve calculations:

Mortality: AM92 Select

Rate of interest: 4% per annum

Expenses: Nil

(c) Explain clearly and fully the reason for the profit or loss that you calculated in (b). [2 marks]

[Total 12 marks]

[**PLEASE TURN OVER]**

Question 5

Using ELT15 (Females), calculate the following:

(a) ${}_{2.5}q_{100.1}$, assuming a uniform distribution of deaths between integer ages.
[2 marks]

(b) ${}_{2.5}q_{100.1}$, assuming a constant force of mortality between integer ages.
[2 marks]

(c) $\mu_{100.1}$, assuming a constant force of mortality between integer ages.
[2 marks]

(d) Comment on your answer to (c). [1 mark]

[Total 7 marks]

Question 6

An endowment assurance policy sold to a man aged 35 pays a benefit after 20 years, or immediately on earlier death. The benefit is calculated as (t pounds) if it is payable at time t years, where t is measured exactly.

(a) Write down the standard actuarial symbol for the expected present value at age 35 of this benefit.
[1 mark]

(b) Calculate the value of this symbol, using the following basis:

Mortality: $\mu_x = 0.0008$ at all ages

Force of interest: 1% per year [7 marks]

(c) Comment on your answer to (b). [2 marks]

[Total 10 marks]

[**PLEASE TURN OVER]**

Question 7

A life office sells a special 25-year with-profits endowment assurance policy to a life aged 40. The basic sum assured is £200,000, payable on maturity at age 65, or at the end of the year of earlier death. A bonus is paid on maturity, but no bonus is paid on death.

Level premiums are paid monthly in advance for a maximum of 20 years (240 payments). They stop at age 60 (not 65), or on the policyholder's death if earlier.

(a) The company calculates premiums on the following basis:

Mortality: AM92 select

Rate of interest: 4% per year

Initial expenses: £600, plus 80% of one year's premium

Renewal expenses: £80 at the start of each year, not including the first year, plus 2% of each premium (including the first). *Renewal expenses stop when the premiums stop.*

Claim expenses: £100 on death, no expenses on maturity.

Bonus: 60% of the sum assured.

Calculate the monthly premium. [8 marks]

(b) A student says that it is not fair that no bonus is paid on death. Discuss this suggestion. [2 marks]

[Total 10 marks]

[Note: the phrase in italics was not in the original exam. The exam question was ambiguous and marks were awarded for an alternative interpretation.]

END OF PAPER